Study Title: Interactions with large language models to promote positive attitudes toward refugees

1) Has any data been collected for this study already?
No, no data for the pre-registered confirmatory study has been collected yet.

2) What's the main question being asked or hypothesis being tested in this study?
The study aims to examine whether interactions with an intervention chatbot lead to more positive attitudes towards refugees compared to interacting with a neutral chatbot.
Attitudes are a multi-dimensional construct. We distinguish three primary dependent variables and test each in a separate model, with one primary hypothesis per dimension.

Primary Hypotheses:
Participants interacting with one of the treatment chatbots will show higher post-intervention attitude scores compared to those in the control group (= a neutral chatbot).
Any comparisons between the different treatment chatbot intervention conditions will be exploratory.
H1 (Perceptions): Treatment > Control
H2 (Refugee support): Treatment > Control
H3 (Policy endorsement): Treatment > Control

3) Describe the key dependent variable(s) specifying how they will be measured.
The primary dependent variable is participants' post-intervention attitudes toward refugees, assessed immediately after the chatbot interaction. We measure using a multi-item survey consisting of three dimensions:

- Perceptions: cognitive and affective evaluations of refugees, adapted from the asylum-seeker component of Hangartner et al. (2019).
- Refugee support: emotional concern and perceived responsibility toward refugees, drawing on items used by Czarnek et al. (2024).
- Policy endorsement: support for refugee-related policies, adapted from measures used in Vlasceanu et al. (2024).

Items are answered on Likert-type scales or 0-100 sliders. Each dimension will be answered through a separate regression analysis. For each dimension, we hypothesize that participants in the treatment-bot conditions will score higher than those in the neutral bot condition.
A two-week follow-up will be conducted to assess the durability of effects.

4) How many and which conditions will participants be assigned to?
Participants will be randomly assigned to one of six conditions in a between-subjects design. Four of these conditions are the treatment groups, in which participants interact with different LLM-powered chatbot interventions.
Treatment groups:

- Factual Debunking Chatbot: addresses common misconceptions about refugees by providing fact-based responses.
- Self-Reflection Chatbot: guides participants through self-reflection and internal reasoning without introducing new factual information.

- Empathy Chatbot: designed to increase empathy toward refugees by encouraging perspective-taking, emotional engagement, and reflection on refugees' experiences.
- Roleplay Chatbot: designed to bring participants into the perspective of a refugee through an empathetic, first-person conversational roleplay.

The other two conditions serve as control groups.

Control group:

- Neutral Chatbot: guides a conversation unrelated to refugees, which thus controls for the general experience of engaging with a chatbot.

Additional comparison group:

- Static Information Condition: a written, non-interactive summary about refugees, controlling for the interaction format (i.e., exposure to information without interaction).

5) Specify exactly which analyses you will conduct to examine the main question/hypothesis.

We will analyze each DV using linear regression models, comparing each treatment condition to the neutral bot as reference category. Confirmatory tests will be conducted using Holm-corrected pairwise comparisons ($\alpha = 0.05$). Exploratory analysis will consist of additional contrasts within the same regression framework.

6) Describe exactly how outliers will be defined and handled, and your precise rule(s) for excluding observations.

Participants will be excluded from the analysis if they fail attention checks, which are embedded within the surveys, complete the study unusually quickly (less than one-third of the median completion time), or provide incoherent or low-effort responses, such as very short answers (fewer than three words) or placeholder/dummy text. Additional exclusions will apply to participants who encounter technical issues during their interaction with the LLM-powered chatbot. The number of participants excluded under each criterion, along with the justification for their exclusion, will be reported in the results to ensure transparency and data integrity.

7) How many observations will be collected or what will determine sample size?

The required sample size was determined via a priori sample size calculation via G*Power (version 3.1.9.7), targeting a significance level of 0.05 (Bonferroni corrected = 0.0125), 80% power, and a medium effect size ($d = 0.5$). The analysis suggests a minimum of 468 participants (approximately 78 per condition across six conditions). To account for exclusions and dropouts, the sample size will be increased by 20%. This results in a final recruitment goal of approximately 562 participants (about 94 per condition).

Participants will be recruited through Prolific, targeting a diverse sample of adults residing in the United Kingdom. Inclusion criteria are: minimum age of 18, English as a first language, an approval rate of at least 98%, and at least 50 previous submissions.

8) Anything else you would like to pre-register?

In addition to the primary analyses, several elements of the study will be documented for transparency. We will collect standard sociodemographic variables. These will not be used in confirmatory analyses but may be examined descriptively or in exploratory subgroup analyses.

9) References

Czarnek, G., Orchinik, L., Rand, D. G., Lin, H., Xu, H. G., Costello, T., Pennycook, G., & Rand, D. G. (2025). *Addressing climate change skepticism and inaction using human-AI dialogues* [Working paper]. OSF. https://osf.io/preprints/psyarxiv/mqcwj_v1

Hangartner, D., Dinas, E., Marbach, M., Matakos, K., & Xefteris, D. (2019). Does exposure to the refugee crisis make natives more hostile? American Political Science Review, 113(2), 442–455. https://doi.org/10.1017/S0003055418000813

Vlasceanu, M., Doell, K. C., Bak-Coleman, J. B., et al. (2024). Addressing climate change with behavioral science: A global intervention tournament in 63 countries. *Science Advances, 10(6)*, eadj5778. https://doi.org/10.1126/sciadv.adj5778